
Amazon QuickSight – Complete Overview

Amazon **QuickSight** is a **serverless, cloud-native business intelligence (BI) service** that lets you create and publish interactive dashboards and visualizations from your data.

What Is QuickSight?

- Fully managed **BI tool** by AWS
 - Used to **visualize data** from multiple sources (S3, Athena, Redshift, RDS, etc.)
 - Supports **ML insights, ad-hoc analysis, scheduled reports, and embedded analytics**
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Key Features of Amazon QuickSight

Feature	Description
SPICE Engine ⚡	In-memory storage engine for fast performance
Dashboards 	Interactive dashboards for visual insights
ML Insights 	Anomaly detection, forecasting
Embedded Analytics 	Integrate dashboards into apps or portals
Ad Hoc Analysis 	Drag-and-drop visual analysis

Multi-Source Support



Connects to over 25 AWS and external sources

Supported Data Sources

AWS Services	3rd Party
S3 (via Athena/Glue)	Salesforce
RDS (MySQL, PostgreSQL, etc.)	SQL Server
Redshift	Excel/CSV uploads
Athena	Snowflake, Teradata
Aurora	JDBC/ODBC

Security Features

Feature	Description
IAM + QuickSight Users	Access control and permissions
Row-Level Security (RLS)	Restrict rows per user/group
KMS Encryption	Data at rest
VPC Connectivity	Private data access from RDS, Redshift

Common Use Cases

- Dashboards for **sales, finance, operations**
 - Reporting from **Redshift or S3 via Athena**
 - Embedded analytics in SaaS products
 - Ad-hoc data exploration
 - **Data-driven decisions** for enterprise teams
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QuickSight Editions

Edition	Features
Standard	Personal dashboards, limited sources
Enterprise	IAM, ML insights, RLS, SPICE, embedded BI
Q (optional)	Ask data questions in natural language

Step-by-Step Lab: S3 + Athena + QuickSight

Objective:

Use a CSV file in S3, query it with Athena, and build a dashboard in QuickSight.

Step 1: Upload Data to S3

1. Go to S3 → Create a bucket (e.g., **quicksight-lab-data**)

2. Upload a CSV file ([sales.csv](#)) like:

```
order_id,product,category,amount,region
1001,Laptop,Electronics,75000,South
1002,TV,Electronics,50000,North
1003,Shoes,Fashion,3000,East
```

✓ Step 2: Crawl the Data with AWS Glue

1. Go to AWS Glue → Crawlers → Add crawler
 2. Choose S3 path: [s3://quicksight-lab-data/](#)
 3. Create a new database (e.g., [qsdb](#))
 4. Run crawler to create a table in Glue Data Catalog
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✓ Step 3: Query Data Using Athena

1. Go to Athena → Choose database [qsdb](#)
 2. Run: [SELECT * FROM sales_csv_table;](#)
 3. Verify the results and save the query
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✓ Step 4: Setup Amazon QuickSight

1. Open QuickSight (<https://quicksight.aws.amazon.com/>)
 2. Sign up and choose **Enterprise Edition**
 3. Configure:
 - Enable Athena & S3 access
 - Add S3 bucket permission: **quicksight-lab-data**
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✅ Step 5: Create Data Set in QuickSight

1. Choose **Athena** as data source
 2. Select your database and table (from Glue)
 3. Import data into SPICE (faster) or use direct query
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✅ Step 6: Create Dashboard

1. Click **New Analysis**
2. Choose your dataset
3. Add visuals:
 - Bar chart: Category vs Amount
 - Pie chart: Region-wise sales

- KPI: Total sales

4. Customize titles, colors, filters

✅ Step 7: Share and Publish

1. Publish dashboard
2. Share with other QuickSight users or groups
3. (Optional) Embed in app or website using API

Sample Visual Ideas

-  Bar: Product Category vs Sales
-  Pie: Sales Distribution by Region
-  KPI: Total Monthly Revenue
-  Line: Revenue Trend Over Time

Pricing Summary

Component	Charges
SPICE	\$0.25 per GB/month
Storage	

Author	\$24/user/month
Reader	\$0.30/session or \$5/user/month
Q (Optional)	\$10/user/month

QuickSight vs Other BI Tools

Tool	QuickSight	Tableau	Power BI
Cloud-Native	✓	✗	✗
Serverless	✓	✗	✗
Embedded Analytics	✓	✓	✗
Pay-per-session	✓	✗	✗

Summary

-  Connects to S3, Redshift, Athena, RDS, etc.
 -  Build powerful dashboards without managing infrastructure
 -  Secure with IAM, RLS, and VPC support
 -  Powered by SPICE for lightning-fast queries
 -  Built-in ML insights + Q for natural language querying
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